

**PEDESTRIAN DETECTION AND
SIGNAL ADJUSTMENT**

Group 25-26J-473

Project Proposal Report

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Declaration page of the candidates & supervisor

I declare that this is my own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Abstract

Pedestrian safety is a cornerstone of sustainable and inclusive urban transportation systems. While significant advances have been made in intelligent traffic signal control through artificial intelligence (AI) and reinforcement learning, the pedestrian component often remains underrepresented. Conventional pedestrian signals use fixed timers that do not adapt to real-world variations in pedestrian mobility, such as the slower walking speeds of elderly individuals or the unique requirements of differently abled pedestrians.

This research proposes a pedestrian group classification and adaptive signal adjustment system that integrates computer vision techniques (YOLO-based detection) with the SUMO (Simulation of Urban Mobility) platform. Pedestrians will be classified into age-related categories—children, adults, and elderly—each modeled with distinct mobility characteristics. The system dynamically adjusts crossing times through TraCI (Traffic Control Interface) based on real-time pedestrian counts and group composition.

Additionally, assistive technologies such as visual countdown timers and auditory crossing cues will be simulated to enhance inclusivity. The system is also designed to respond to emergency vehicles by pausing pedestrian signals to ensure rapid clearance. Evaluation will be conducted by comparing the adaptive system with conventional fixed pedestrian signal control, analyzing metrics such as pedestrian wait time, crossing safety, and vehicle delay.

The novelty of this approach lies in its human-centric perspective, prioritizing vulnerable groups and ensuring accessibility, thereby aligning with broader smart city goals

Keywords - YOLO (You Only Look Once), Deep learning, Audio-based detection, Siren sound recognition, Indicator detection, HSV colour space segmentation, Flashing light bar detection

List Of Abbreviations

Abbreviation	Description
YOLO	You Only Look Once (Object Detection Model)
SUMO	Simulation of Urban MObility
TraCI	Traffic Control Interface
AI	Artificial Intelligence
ML	Machine Learning
LKR	Sri Lankan Rupees

Table 1 : List of Abbreviations

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1. Introduction

1.1. Background and Literature survey

Pedestrian safety has become one of the most critical challenges in the context of modern Intelligent Transportation Systems (ITS) and smart city development. With rapid urbanization, intersections are increasingly shared by diverse groups of pedestrians—children, adults, elderly individuals, and differently-abled persons—each possessing distinct mobility patterns and walking speeds. Conventional fixed-time pedestrian signals fail to account for this variability, often leading to unsafe crossings, increased wait times, and accessibility issues for vulnerable groups [1].

Research on adaptive traffic signal control has primarily emphasized vehicle-centric optimization, with the aim of reducing vehicle delays and congestion [2]. However, pedestrian behavior is typically modeled in a simplistic way, with uniform assumptions about walking speed and crossing patterns [3]. This approach overlooks critical real-world variability, such as slower speeds among elderly pedestrians or the need for assistive technologies for visually impaired individuals. Such limitations highlight the urgent need for human-centered adaptive signal systems that balance safety, accessibility, and traffic efficiency.

Recent advances in computer vision and deep learning (e.g., YOLO, Faster R-CNN) have enabled more accurate pedestrian detection and classification in real-time [4]. Studies have shown that integrating these detection methods into traffic simulators such as SUMO (Simulation of Urban Mobility) allows for the modeling of heterogeneous pedestrian behavior, including differentiated crossing speeds and dynamic pedestrian flow [5]. Moreover, assistive signal technologies, such as audio guidance, haptic alerts, and visual countdown timers, have been identified as key enablers for enhancing the inclusivity of urban mobility systems [6].

While some research has investigated adaptive pedestrian signals, the majority of studies focus only on crowd size without distinguishing among age groups or differently-abled pedestrians [7]. Few systems integrate real-time pedestrian classification with signal timing adjustments and even fewer combine these with emergency-awareness features that pause pedestrian signals when ambulances or fire trucks are approaching [8].

The proposed component addresses these gaps by introducing a pedestrian group classification and adaptive signal adjustment module. By leveraging computer vision (YOLO-based models) for pedestrian type recognition, combined with SUMO for simulation, the system dynamically adjusts crossing durations based on real-time pedestrian characteristics. Additionally, it integrates assistive technologies (audio-visual cues) to improve accessibility and inclusivity, ensuring a safer and smarter pedestrian crossing environment that aligns with global sustainable urban mobility goals.

Advantages of the Proposed System

- Inclusivity – Ensures safe crossing for elderly and differently abled pedestrians.
- Adaptivity – Dynamically adjusts pedestrian green time based on detected group composition and walking speeds.
- Assistive Features – Provides visual countdowns and audio cues to improve accessibility.
- Emergency Awareness – Suspends pedestrian phases when emergency vehicles are detected.
- Scalability – Modular system design allows deployment in real-world smart intersections.
- Novelty – Bridges the gap between computer vision detection models and SUMO-based adaptive pedestrian signal control.

1.2. Research Gap

Traditional pedestrian signal systems operate on fixed-time control, where green phases are predetermined and remain unchanged regardless of the actual pedestrian presence or mobility needs. Such systems often disadvantage vulnerable groups such as elderly individuals, children, or differently-abled pedestrians, who may require more time to cross safely. Moreover, traditional approaches lack real-time detection of pedestrians, relying instead on manual push buttons, which are not inclusive for visually impaired users and are often ignored by pedestrians.

Recent research has introduced sensor-based pedestrian detection (infrared, pressure mats, or camera-based systems). While these improve on fixed timers, they typically treat pedestrians as a homogeneous group, failing to account for age-related walking speeds or behavioral differences. Other systems that integrate computer vision often emphasize vehicle detection and traffic flow optimization, leaving pedestrian inclusivity underexplored.

The proposed Pedestrian Group Classification and Signal Adjustment component addresses these gaps by integrating:

- YOLO-based pedestrian detection to classify children, adults, and elderly.
- SUMO integration with TraCI for dynamic adjustment of crossing time based on pedestrian type.
- Assistive technologies such as audio cues and visual countdown timers for inclusivity.
- Emergency-awareness, pausing pedestrian signals during emergency vehicle prioritization.

This integrated, human-centered approach not only enhances safety but also ensures equal accessibility, which is missing from conventional systems.

Method	Strengths	Research Gaps	References
Fixed-Time Pedestrian Signals	Simple to implement, low infrastructure cost; widely used in most cities.	✗ Does not adapt to real-time pedestrian presence. ✗ Ignores diverse walking speeds of children/elderly. ✗ Leads to inefficiency during low/high pedestrian flow.	[1], [2]
Crowd-Size Based Adaptive Signals	Adjusts green time based on pedestrian count; reduces waiting time during peak hours.	✗ Does not differentiate pedestrian groups (all assumed same speed). ✗ Fails to account for accessibility of elderly or disabled individuals.	[3], [4]
Computer Vision (YOLO/Faster R-CNN)	High accuracy in pedestrian detection and classification; real-time performance possible.	✗ Requires large, diverse datasets (children, elderly, disabled often under-represented). ✗ Sensitive to lighting/weather variations.	[5], [6]
Wearable/Smartphone-Based Systems	Provides personalized crossing assistance (alerts, navigation).	✗ Requires active participation from	[7], [8]

Method	Strengths	Research Gaps	References
		pedestrians (not scalable). ✗ Limited adoption among elderly/disabled groups.	
Assistive Crossing Technologies (Countdown timers, Audio Cues)	Improves inclusivity for visually/physically impaired pedestrians.	✗ Works only when signal durations are already optimized. ✗ Does not dynamically adapt to pedestrian type in real-time.	[9], [10]
Hybrid Adaptive Systems (Vision + Assistive Features + SUMO Simulation)	Combines real-time pedestrian detection with adaptive signal timing and inclusive assistive technologies; supports simulation/testing.	✗ Still in early research phase; requires robust deployment strategy in real-world smart cities.	[11], [12]

Table 2: Competitive Analysis

1.3. Research Problem

In modern urban environments, pedestrian safety and accessibility remain critical challenges within traffic management systems. Conventional pedestrian signal systems typically operate on fixed timers or basic push-button triggers, lacking the ability to dynamically adapt to the diverse characteristics of pedestrians. While some systems incorporate crowd-size based adjustments, they fail to distinguish among different pedestrian groups such as children, elderly, and differently abled individuals, each of whom have distinct walking speeds and safety requirements.

Existing approaches often rely on single-parameter adjustments (e.g., pedestrian count, estimated flow), which limits their robustness and inclusivity. For example, uniform green signal durations may be sufficient for adults but insufficient for elderly or mobility-impaired pedestrians, leading to increased safety risks at crossings. Similarly, the absence of assistive features such as visual countdowns or audio guidance reduces accessibility for vulnerable groups.

Furthermore, current pedestrian signal studies are typically evaluated in isolated test environments without integration into realistic urban traffic simulations such as SUMO. This prevents accurate analysis of trade-offs between pedestrian safety, wait times, and vehicle delays. The lack of multi-faceted modeling—combining pedestrian group classification, adaptive signal adjustment, and assistive crossing technologies—creates a significant gap in deploying effective and inclusive smart traffic solutions.

The proposed Pedestrian Group Classification and Signal Adjustment component aims to bridge this gap by introducing a multi-modal, human-centered solution that dynamically adjusts signal timings based on real-time pedestrian classification and presence. Integrated into SUMO for simulation and evaluation, the system offers the potential to improve both safety and accessibility while aligning with broader smart city sustainability goals.

2. Objectives

2.1. Main Objective

To design and implement an adaptive pedestrian signal control system that classifies pedestrian groups and adjusts crossing times dynamically, integrating inclusivity and emergency-awareness.

2.2. Specific Objectives

- Model pedestrian groups (children, adults, elderly) with realistic walking speeds in SUMO.
- Train a YOLO-based pedestrian detection model capable of classifying pedestrians by group.
- Implement real-time integration of pedestrian counts and group types into SUMO via TraCI.
- Simulate assistive features like visual countdowns and audio crossing cues.
- Incorporate emergency vehicle detection to pause pedestrian phases when needed.
- Evaluate system performance against fixed pedestrian signals using pedestrian wait time, safety rate, and vehicle delay.

3. Methodology

The methodology defines the approach to designing, developing, and evaluating the proposed pedestrian group classification and signal adjustment system. It combines computer vision (YOLO), SUMO simulation, and assistive technologies to create a comprehensive human-centered traffic control system.

3.1. Requirement Gathering

Requirement gathering for this component involves collecting, organizing, and preparing all necessary datasets, system specifications, and functional needs to ensure accurate Pedestrian detection and effective traffic signal pre-emption within the SUMO simulation environment.

3.1.1. Dataset Collection

To achieve robust detection under varied conditions, a multi-source dataset acquisition strategy will be used:

- Source: Real-world pedestrian video recordings (collected ethically from public intersections using a mobile device at a distance).
- Categories: Children, adults, elderly, and differently-abled individuals.

- Preprocessing: Frame sampling, annotation using LabelImg/Roboflow, and dataset augmentation (rotation, brightness, scaling).
- Additional Datasets: Publicly available pedestrian datasets such as INRIA, Caltech Pedestrian, and CrowdHuman.

3.2. Development Methodology

3.2.1. Visual Detection Module

Description:

Implement a **YOLO-based real-time pedestrian detection and grouping module** integrated with **OpenCV** to identify pedestrians from live or simulated camera feeds (and SUMO-rendered videos). Beyond detecting “person,” the module **classifies pedestrians into groups**—*children, adults, elderly* (and optionally *mobility-assisted* like cane/walker/wheelchair if labeled)—to inform **adaptive crossing time**. The detector will be **fine-tuned** on a curated dataset spanning **day/night, rain/fog, occlusions, crowds, and different camera angles/heights**.

Core Tasks:

- Detect pedestrians at ≥ 30 FPS (edge GPU) with **YOLOv8/YOLOv5**.
- **Multi-class labeling**: map detections to *child/adult/elderly* using either (a) a single multi-class YOLO head or (b) YOLO person detection + a **second-stage classifier** (lightweight CNN) on cropped person images.
- Estimate **walking speed proxy** (bbox displacement over time) to refine timing beyond static class.
- Publish counts & group mix (e.g., {children: 3, adults: 8, elderly: 2}) at **1–2 Hz** to the Signal Controller.

Inputs / Outputs:

- **Input**: RTSP/USB camera feed (or SUMO video), optional ROI masks (crosswalk polygons).
- **Output (per tick)**:

- counts_by_group (children/adults/elderly)
- avg_speed_by_group (px/s → mapped to m/s if calibrated)
- crosswalk_occupancy (boolean, density %)
- confidence_stats (mean confidence, #low-conf detections)

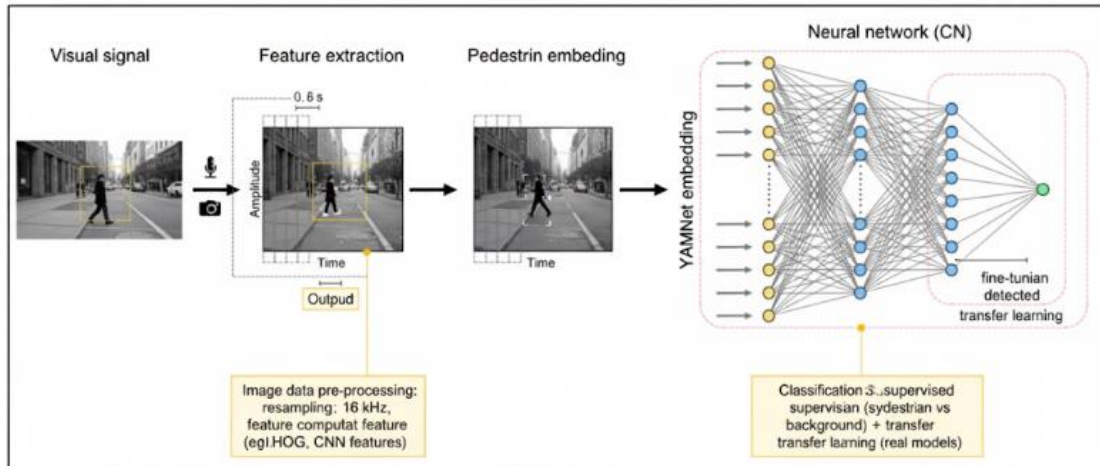


Figure 1 : Pedestrian Detection Source [14]

3.2.2. SUMO and TraCI Integration

This module connects the **pedestrian detection system** with the **SUMO traffic simulation environment** through the **TraCI API** to dynamically adjust pedestrian crossing phases in real-time. When the visual detection module identifies pedestrians (classified into children, adults, and elderly), their presence and estimated walking speeds are sent to SUMO. Using TraCI, the simulation adapts **signal timing** (green/red phases) based on group composition—e.g., granting longer green times when elderly or mobility-impaired pedestrians are detected, or shortening when only fast-moving adults are present. This ensures the simulation reflects **human-centric adaptive control** rather than static rules.

The SUMO–TraCI link allows:

- **Dynamic pedestrian green phase extension/reduction** based on detection results.
- **Priority balancing** between pedestrian safety and vehicle throughput.

- **Integration of assistive features** (countdown timers, audio signals) within the simulation.
- **Emergency-awareness:** pause/delay pedestrian signals if emergency vehicle priority is triggered by another module.
- **Real-time data flow** from YOLO → Python middleware → SUMO (via TraCI).

Technologies & Tools:

- **SUMO (Simulation of Urban Mobility):** Base simulation engine for road networks, pedestrian crossings, and traffic signals.
- **TraCI (Traffic Control Interface):** Python API to control signal phases, retrieve pedestrian/vehicle states, and integrate external models.
- **NetEdit:** SUMO's network editor to design realistic intersections, crosswalks, and signal plans.
- **Python Middleware:** Collects YOLO detection results, translates them into **green time adjustments**, and communicates with SUMO.
- **Matplotlib / SUMO-GUI (optional):** For visual validation and logging adaptive signal performance.

Illustration:

Figure 2 shows the **virtual environment in SUMO**, where intersections, signals, and pedestrian crossings can be modeled. This flexible setup allows portable, reproducible experiments to evaluate adaptive pedestrian signal strategies before real-world deployment.

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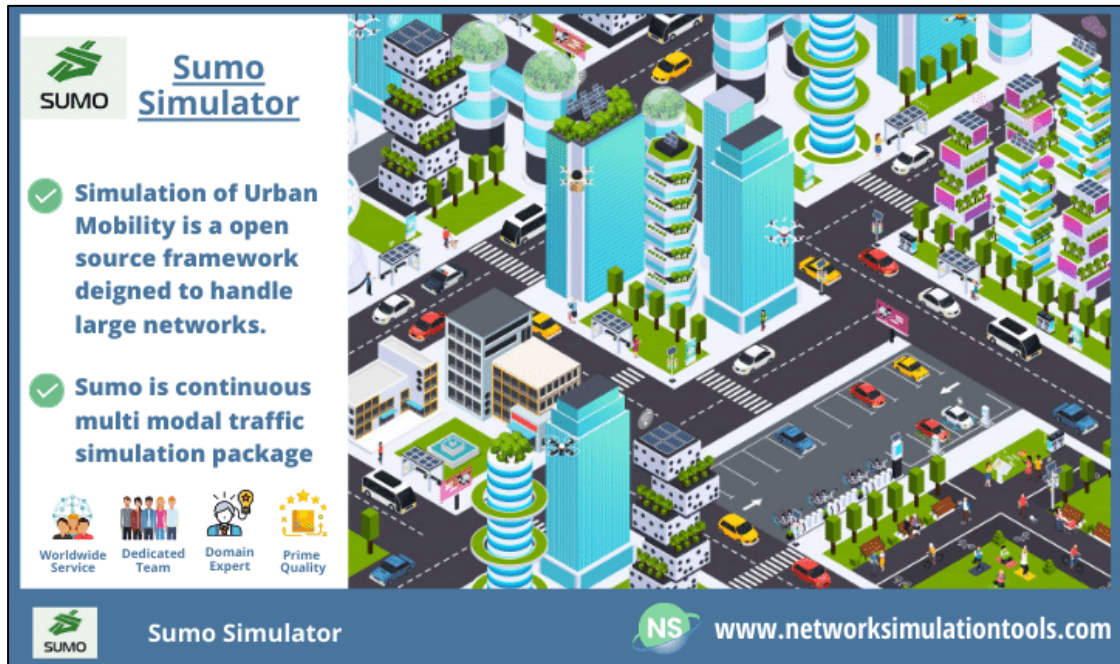


Figure 2 : SUMO Stimulator , Source : <https://networksimulationtools.com/sumo-simulator/>

3.2.3. Dashboard Visualization & Data Sharing

Description:

Develop a **real-time dashboard** to visualize pedestrian detection results, signal control decisions, and system performance metrics. The dashboard will present:

- **Pedestrian detection data:** Number of children, adults, and elderly detected per crossing cycle.
- **Adaptive signal actions:** Current traffic light phase and green-time adjustments applied.
- **Assistive feature indicators:** Status of visual countdowns and audio guidance triggers.
- **System performance metrics:** Average pedestrian wait time, crossing completion rate, and vehicle delay statistics.
- **Alerts & events:** Notifications when vulnerable pedestrians (e.g., elderly, differently-abled) are present, or when pedestrian signals are temporarily paused for emergency vehicle pre-emption.

Additionally, the dashboard will **log and transmit summarized data** to relevant traffic authorities or smart city monitoring platforms for evaluation and long-term policy planning.

Technologies & Tools:

- **Flask / FastAPI (Python):** Backend API to serve pedestrian detection outputs, SUMO simulation data, and adaptive signal decisions.
- **React.js:** Frontend dashboard for **real-time visualization** (charts, tables, alerts, traffic light status).

3.2.4. System Performance Evaluation

Description:

The performance of the **Pedestrian Group Classification and Signal Adjustment system** will be rigorously evaluated within the SUMO simulation environment under varying conditions such as **traffic density**, **pedestrian flow volumes**, **lighting/weather conditions** (day/night, fog, rain), and **crowd diversity** (different proportions of children, adults, and elderly).

The evaluation will focus on both **technical performance** (accuracy, latency, reliability) and **functional outcomes** (pedestrian safety, accessibility, and vehicle delay trade-offs). Specific metrics include:

- **Detection accuracy:** Correct classification of pedestrians into children/adult/elderly categories.
- **False positives/negatives:** Incorrect or missed classifications of pedestrian groups.
- **System latency:** Time from video frame capture → YOLO detection → TraCI green-time adjustment.
- **Pedestrian safety outcomes:** Successful crossing completion rates, reduction in mid-crossing conflicts.
- **Pedestrian wait times:** Average waiting duration before green signal.

- **Vehicle delay impact:** Extra vehicle waiting time introduced due to longer pedestrian phases.
- **Comparison against baseline:** Benchmark against fixed pedestrian signal systems to measure improvements.

Technologies & Tools:

- **Python (NumPy, Pandas):** For simulation log parsing, data processing, and statistical evaluation.
- **Matplotlib / Seaborn:** For visual representation of performance metrics (line graphs, bar charts, heatmaps).
- **SUMO + TraCI Logs:** To collect real-time pedestrian and signal state data.
- **Jupyter Notebook / Google Colab:** For structured evaluation and reproducibility.

Evaluation Approach:

1. **Scenario definition:** Configure multiple SUMO intersections with varying pedestrian flows and vehicle densities.
2. **Simulation runs:** Execute repeated trials (e.g., 30 runs per condition) with random seeds for statistical reliability.
3. **Performance logging:** Store detection outputs, signal phase changes, pedestrian counts, and vehicle delays.
4. **Statistical analysis:** Compute averages, standard deviations, and confidence intervals for each metric.
5. **Visualization:** Generate comparative graphs to highlight improvements over fixed-timer systems.
6. **Reporting:** Summarize insights into dashboards and research findings for continuous system refinement.

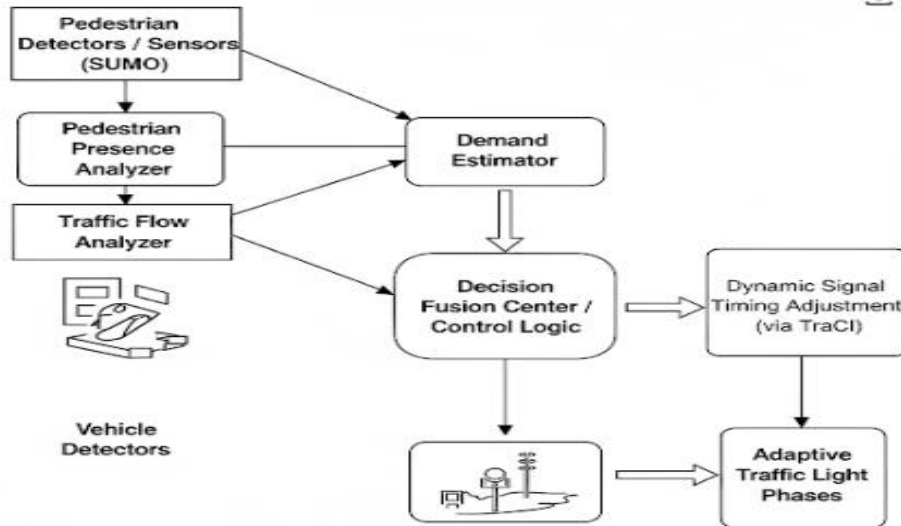


Figure 3 :Component Flow Diagram

3.3. Project Management Methodology

The development and evaluation process will follow the **Agile software development methodology**, chosen due to the **complexity** of integrating multi-modal detection (YOLO classification, SUMO simulation, assistive technologies) and the need for **iterative improvements**. Agile enables:

- Incremental progress with continuous testing in SUMO.
- Flexibility to refine YOLO models or SUMO scenarios as data evolves.
- Feedback-driven adjustments from evaluation results to improve accuracy and inclusivity.
- Transparent tracking of tasks and deliverables using Agile project management tools (e.g., Jira)

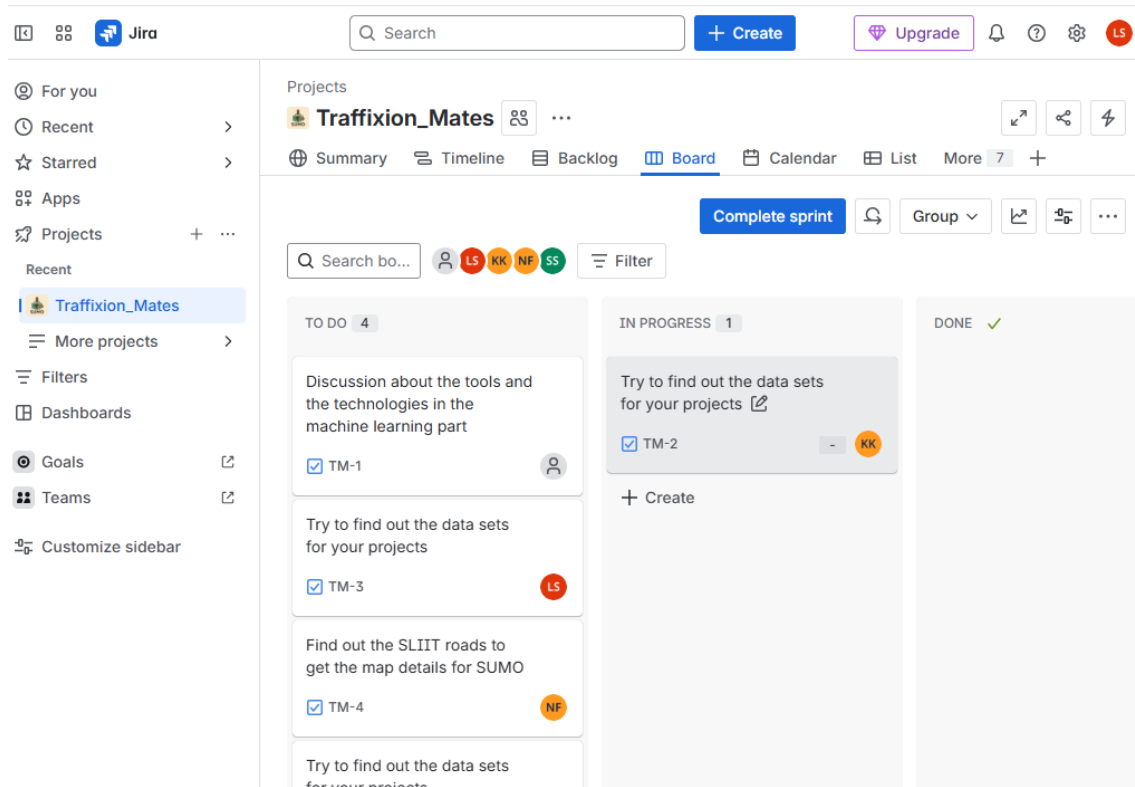


Figure 4 : Jira management tool

3.4. Feasibility Study

The feasibility study evaluates the practicality, resources, and challenges involved in implementing the Pedestrian Group Classification and Adaptive Signal Control system. It considers the technical feasibility, schedule feasibility, and economic viability of deploying the solution as part of a smart city traffic management framework.

3.4.1. Technical Feasibility

The proposed system combines computer vision, machine learning, and traffic simulation technologies to detect pedestrians, classify them by group (children, adults, elderly), and adjust pedestrian crossing signals in real-time. The technical feasibility is supported by the team's access to required tools, open-source frameworks, and domain knowledge.

Knowledge on Technologies

Technology Area	Competencies Required	Applicability in Project
Computer Vision	Proficiency in YOLOv8 / YOLOv5 and OpenCV	Detecting pedestrians in video streams and classifying them into groups.
Machine Learning	Experience in PyTorch / TensorFlow for model training & fine-tuning	Building custom classifiers for children, adults, elderly pedestrians.
Dataset Handling	Familiarity with Labellmg , Roboflow , and augmentation techniques	Annotating pedestrian datasets, handling class imbalance, synthetic data generation.
Traffic Simulation	Hands-on knowledge of SUMO & TraCI API	Controlling signal phases dynamically based on pedestrian detection.
Data Integration	Ability to connect Python middleware with SUMO (TraCI)	Real-time communication between YOLO detections and SUMO signals.
Dashboard Development	Skills in Flask/FastAPI + React.js	Visualization of pedestrian flows, signal timings, and performance metrics.
System Optimization	GPU acceleration via CUDA / cuDNN	Ensuring real-time processing and low latency detection.

Table 4 : Knowledge and technologies

Technical Resources Available:

- Access to GPUs (NVIDIA CUDA-supported) for model training and inference.
- Open-source datasets (Caltech, INRIA, CrowdHuman) supplemented with own-labeled footage.
- SUMO simulation platform with TraCI integration.
- Team experience in Python, PyTorch, and traffic system modeling.

This ensures the system is technically feasible, with minimal dependency on proprietary tools, making it cost-effective and reproducible.

3.4.2. Schedule Feasibility

The project will follow the Agile methodology, divided into 2–3 week sprints for each module:

- Dataset preparation (Sprint 1)
- Visual detection (Sprint 2)
- Acoustic detection (Sprint 3)
- Hue-based detection (Sprint 4)
- SUMO & TraCI integration (Sprint 5)
- Indicator detection (Sprint 6)
- Dashboard development (Sprint 7)
- Testing & optimization (Sprint 8)

Given the modular approach, the timeline is realistic and allows flexibility for refinement based on test feedback. Milestones will be tracked in Jira, ensuring tasks are delivered within deadlines.

3.5. System Analysis

3.5.1. Software Solution Approach

The proposed system will be developed as a modular, multi-modal detection and simulation platform that integrates visual, acoustic, and hue-based detection techniques with the SUMO traffic simulation to enable real-time pedestrian prioritization.

Data Collection -Gather emergency vehicle image datasets (Google, Kaggle, real-world) and siren audio datasets (UrbanSound8K, Kaggle, recordings).

Pre-processing -Augment and normalize images.

Model Training -Use YOLOv8/YOLOv5 for visual detection, HSV + Haar Cascade for light bar detection.

Multi-Modal Fusion -Combine visual, audio, and hue-based outputs via decision fusion to improve accuracy.

SUMO Integration -Control traffic signals dynamically with TraCI when emergency vehicles are detected.

Route Prediction -Detect turn signals to predict paths and share with nearby intersections.

Dashboard -Real-time monitoring via React.js frontend and Flask/FastAPI backend, with charts/maps.

3.5.2. Tools and Technologies

The proposed system will be developed using a combination of software frameworks, programming languages, simulation tools, and development platforms to ensure robustness, scalability, and real-time performance.

Programming Languages

- Python 3.x -The core development language for computer vision, audio processing, machine learning, and SUMO integration due to its extensive library ecosystem.
- JavaScript (React.js) -For building the real-time interactive dashboard frontend

Frameworks and Libraries

- YOLOv8 / YOLOv5 (PyTorch) -State-of-the-art deep learning frameworks for real-time visual emergency vehicle detection.
- OpenCV -Computer vision library for video frame processing, HSV colour segmentation, and Haar Cascade implementation.
- TensorFlow / Keras -Machine learning frameworks for training CNN/LSTM models for siren sound recognition.
- Librosa -Python library for audio feature extraction such as MFCC, used in acoustic detection.
- Scikit-learn -For data preprocessing, model evaluation, and integration of classical ML methods.

Traffic Simulation and Integration Tools

- SUMO (Simulation of Urban Mobility) -Microscopic, open-source traffic simulator for testing signal pre-emption strategies in urban environments.
- TraCI API -Traffic Control Interface for real-time control of SUMO simulations through Python scripts.
- NetEdit -SUMO's network editor for building and modifying traffic network layouts.

Data Collection and Annotation Tools

- Google Images / Kaggle Datasets -Sources for diverse emergency vehicle, siren, and traffic datasets.
- LabelImg / Roboflow -Annotation tools for creating bounding box labels for training YOLO models

Backend Development and API Tools

- Flask / FastAPI -Python-based web frameworks for creating RESTful APIs to serve detection and simulation data to the dashboard.

Frontend Development and Visualization Tools

- React.js -For creating a responsive, real-time web dashboard for visualizing detections and simulation status.

Development and Project Management Tools

- VS Code -Main integrated development environment for coding and debugging.
- Google Colab -For GPU-enabled training of detection and audio models.
- Jira -For Agile sprint planning, backlog management, and task tracking.
- Git & GitHub -For version control and collaborative development.

Testing Tools

- Pytest -For automated unit and integration testing of Python modules.
- Postman -For testing and validating API endpoints.

Hardware Requirements

- GPU-enabled Workstation / Cloud GPU -For training deep learning models efficiently.
- CCTV / IP Cameras -For real-world visual data collection and live feed integration.

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3.6. Project Requirements

The project requirements define the expected functionality and operational qualities of the Pedestrian Movement and Dynamic Signal Control component in SUMO Simulator. These requirements ensure that the system provides realistic pedestrian simulations and adaptive traffic management for safer and more efficient crossings.

3.6.1. Functional Requirements

The system shall:

- **Pedestrian Simulation** – Accurately simulate realistic pedestrian movement, including walking speeds, waiting behavior, and crossing decisions.
- **Real-Time Pedestrian Detection** – Monitor pedestrian presence at crossings in real time using SUMO detectors or TraCI API.
- **Dynamic Signal Control** – Adjust traffic light phases dynamically to allocate crossing time based on pedestrian demand and density.
- **Behavior Modeling** – Incorporate behavioral patterns such as group crossing, impatience, and jaywalking tendencies.
- **Multi-Intersection Coordination** – Share pedestrian density data with neighboring intersections to synchronize crossing times.
- **Dashboard Visualization** – Provide a web-based interface showing pedestrian flows, crossing signals, and simulation status in real time.
- **Data Logging** – Record pedestrian counts, waiting times, signal adjustments, and overall performance metrics for later analysis.
- **Accessibility Considerations** – Simulate extended crossing times for vulnerable pedestrian groups (elderly, disabled).
- **Dashboard Visualization** -Display real-time detection events, simulation updates, and traffic light statuses on a web-based dashboard.

3.6.2. Non-Functional Requirements

The system shall:

- **Performance** – Operate with a decision latency of less than 500 ms for detecting and responding to pedestrian presence.
- **Accuracy** – Correctly adjust crossing times with $\geq 95\%$ accuracy in simulation scenarios.
- **Scalability** – Support large-scale intersections with high pedestrian and vehicle volumes.
- **Reliability** – Maintain consistent performance under varying pedestrian densities and traffic conditions.
- **Security** – Ensure that stored simulation data and logs are protected.
- **Maintainability** – Use modular, well-documented code for easy updates.
- **Usability** – Provide an intuitive dashboard for traffic engineers and researchers.
- **Compatibility** – Run on both local systems and cloud environments with SUMO and TraCI.
- **Resource Efficiency** – Optimize resource usage to run smoothly on mid-range hardware.

3.7. Project Scope

The scope of this project defines the boundaries and extent of the Pedestrian Movement and Dynamic Signal Control in SUMO Simulator, outlining the features and functionalities included in the current phase.

3.7.1. In-Scope

The system will include:

- **Pedestrian Simulation** – Generation of pedestrians with realistic movement patterns.

- **Real-Time Detection** – Monitoring pedestrian presence using detectors and TraCI API.
- **Dynamic Traffic Signal Adjustment** – Adaptive green/red times for pedestrians and vehicles.
- **Pedestrian Behavior Modeling** – Consideration of group crossing and varying speeds.
- **Coordination Across Intersections** – Sharing pedestrian demand data between intersections.
- **Dashboard Visualization** – Web-based dashboard showing traffic lights and pedestrian activity.
- **Data Collection** – Logging of pedestrian wait times, crossing delays, and traffic flow.
- **Simulation Testing** – Evaluation under different pedestrian densities and traffic scenarios.

3.7.2. Out-of-Scope

The system will not include:

- **Real-World Deployment** – No direct control of actual pedestrian signals.
- **Integration with CCTV or Sensors** – Only SUMO-generated pedestrians will be used.
- **Pedestrian-Vehicle Accident Modeling** – Safety analysis beyond waiting/crossing times is excluded.
- **Augmented Reality or Training Features** – No gamification or user-based training.
- **Offline Mode** – The system requires SUMO simulation and TraCI; it will not operate standalone.

3.8. Testing

Testing ensures that the Pedestrian Movement and Dynamic Signal Control system works correctly, efficiently, and realistically in SUMO.

3.8.1. Unit Testing

Purpose:

Verify that each module functions independently.

Scope:

- Pedestrian generation and walking behavior.
- Detection of pedestrian presence at crossings.
- Signal adjustment logic for varying pedestrian demand.
- Data logging module (accuracy of saved metrics).

Tools:

- pytest for automated module testing.
- Mock pedestrian datasets

3.8.2. Integration Testing

Purpose:

Ensure that modules work correctly when combined.

Scope:

- Interaction between pedestrian detection and signal control.
- Communication with SUMO using TraCI API.
- Dashboard synchronization with simulation updates.

- Data logging during live simulations.

Tools:

- SUMO TraCI logs.
- Postman (if REST API is used for the dashboard).

3.8.3. Performance Testing

Purpose:

Assess responsiveness and efficiency.

Metrics:

- **Latency:** Time taken to adjust signals after pedestrian detection.
- **Accuracy:** Percentage of correct adjustments compared to ground truth.
- **Throughput:** Number of simulation steps handled per second.
- **Resource Utilization:** CPU, GPU, and memory usage.

Tools:

- Python profiling tools.
- System monitoring utilities.

4. Expected Outcomes

The Emergency Vehicle Detection and Triggering Signal in SUMO Simulator component is expected to deliver measurable improvements in emergency response efficiency through accurate detection, fast decision-making, and intelligent traffic control. The outcomes will focus on performance, usability, and impact within simulated urban traffic environments.

Measurable Performance Goals

High Detection Accuracy

- Achieve $\geq 95\%$ accuracy in detecting pedestrians using multi-modal fusion.
- Reduce false positives and false negatives to $< 5\%$ during simulation tests.

Low Detection Latency

- Process visual and audio input streams with an average latency below 500 ms.
- Maintain real-time frame processing speed of ≥ 20 FPS under GPU-accelerated execution.

Functional Outcomes

- A fully functional YOLO-based real-time visual detection module integrated with OpenCV.

5. Budget and Budget Justification

The cost of development will remain low, as the system primarily uses open-source frameworks and tools. The main expenses will involve data collection equipment and hosting services.

Estimated Budget:

Type	Cost (LKR)	Remarks
GPU-enabled computing (Cloud/Colab Pro)	8,000	Training model fine-tuning
Data Collection (camera)	12000	Real-world capturing
Internets and hosting	6,000	Hosting dashboard
Stationery and documentation	1,000	Printing and report costs
Total	27000	-

Table 3 : Budget and Budget Justification

The low-cost structure, combined with the scalability of the system, makes the project economically viable. The solution can later be commercialized or integrated into existing Intelligent Transportation Systems (ITS) for real-world deployment.

6. Timeline

The project will proceed through a well-organized development timeline. At the outset, the team will concentrate on completing the platform's design, including creating a comprehensive system diagram and choosing the essential technologies. Next, the implementation phase will commence, focusing on fine-tuning and developing algorithms for SUMO Stimulation. Simultaneously, performance analysis tools and other system components will be integrated. The final phase will include extensive testing, performance assessment, and the formulation of commercialization plans. Throughout these stages, the project will emphasize careful development and validation before the platform's official launch.

Task	2025						2026				
	July	August	September	October	November	December	January	February	March	April	May
Feasibility study											
Evaluate feasibility and background study											
Requirement gathering											
Equipment gathering and testing											
Background survey											
Literature review											
Requirement analysis											
Software requirement specification											
Functional, non-functional requirement											
Proposal presentation											
Project proposal report											
Data collection											
ML model creation											
Model training											
Model tuning											
Designing wireframes											
SUMO and Software integration											
Deployment & maintenance											
Progress presentation 1											
Research paper											
Progress presentation 2											
Testing											
Final presentation and viva											

Figure 5 : Component Timeline

7. Communication Management Plan

The Communications Management Plan is designed to ensure that all team members, as well as the supervisor and co-supervisor, receive the necessary information to effectively perform their roles throughout the project. Successful project execution relies on strategic planning and effective communication. This plan outlines how to communicate with

various stakeholders efficiently, detailing the audience, content, format, frequency, and expected outcomes of communications. It also specifies each stakeholder's role, task assignment, and communication strategy based on their influence, interests, and expectations.

Communication Objectives

Proactive communication is essential for project success. Communication should be:

- Adequate: Providing information in the right format and content.
- Specific: Tailored to the targeted audience.
- Sufficient: Including all necessary details.
- Concise: Clear and to the point, avoiding unnecessary repetition.
- Timely: Addressing relevant points at appropriate times.

Communication Media The following media will be utilized for communication throughout the project:

1. Microsoft Teams : For formal updates, documentation, and detailed communications.
2. Documents (MS Word and PowerPoint): For reports, presentations, and detailed project documentation.
3. Phone Calls: For quick discussions and urgent matters.
4. Meetings: Conducted via meeting rooms, conference phones, or Microsoft Teams for in-depth discussions, planning, and progress reviews.
5. Chats (WhatsApp): For informal, quick exchanges and real-time updates.

8. Commercialization

The *Pedestrian Group Classification and Signal Adjustment* component has strong commercialization potential across both public and private sectors. The system addresses a critical gap in pedestrian safety, inclusivity, and accessibility within urban traffic management, aligning with Smart City and Sustainable Mobility goals. By integrating computer vision, adaptive traffic signals, and assistive technologies, it

provides a market-ready solution for improving pedestrian experiences while balancing vehicular flow.

Potential Markets

- **Government Traffic Management Authorities**
City councils and urban planning departments can deploy the system in traffic control centers to reduce accidents, improve safety for elderly and differently-abled pedestrians, and enhance overall urban mobility.
- **Smart City Solution Providers**
Companies offering IoT-based traffic management platforms can integrate the pedestrian detection and adaptive signal module as a premium feature, enhancing their smart intersection offerings.
- **Public Transportation Hubs**
Train stations, bus terminals, and airports can adopt this system to ensure safe pedestrian crossings in high foot-traffic zones, improving flow efficiency and safety.
- **Educational and Healthcare Institutions**
Universities, schools, and hospitals—locations with a higher density of children and elderly individuals—can integrate the system at surrounding intersections to minimize risks and support accessibility.
- **Assistive Technology Companies**
Vendors focusing on accessibility solutions for the visually or physically impaired can partner to incorporate the system’s audio-visual cues into broader inclusive mobility offerings.

Future Commercial Potential

The technology can be extended beyond emergency vehicles to support:

- **Public Transport Priority:** Buses and trams receiving traffic signal priority during peak hours.
- **Autonomous Vehicle Integration:** Providing signal priority for driverless emergency and service vehicles.

9. Summary

The proposed Pedestrian Movement and Dynamic Signal Control in SUMO Simulator component is an intelligent traffic management solution designed to improve pedestrian safety and efficiency in urban environments. By simulating realistic pedestrian behavior and integrating real-time detection through SUMO and TraCI, the system dynamically adjusts traffic signal phases based on pedestrian presence and demand. This ensures optimized crossing times, reduced waiting periods, and better coordination between pedestrians and vehicles.

The project adopts an Agile development methodology, enabling iterative improvements through structured testing phases including unit, integration, simulation, and performance testing. Expected outcomes include reduced pedestrian waiting times, improved traffic flow at intersections, and enhanced safety for vulnerable groups. Additionally, the system provides a scalable and modular foundation that can be extended to larger Intelligent Transportation Systems (ITS) or integrated with real-world sensor data in future phases. With strong potential for smart city applications, the system contributes a cost-effective and impactful solution to modern urban mobility management.

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